

Batch No: 3EC19

20EC280 – Mini Project II

Smart Portable Real Time Detection System for Microplastics in Waterbodies

SDG GOALS MAPPED Nos. : 6,12

Type of Project: Product

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**Project viva-vose Examination
Date : 15.04.2026**

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- Current detection techniques such as FTIR and Raman spectroscopy are highly accurate but expensive, time-consuming, and limited to laboratory use.
- There is a pressing need for a low-cost, portable, and real-time system capable of detecting and quantifying microplastics in diverse water sources efficiently.
- Microplastics, plastic fragments which range from 100 to 300 micrometer, have become a major environmental pollutant, contaminating oceans, rivers, and even drinking water.

Experiment	Inference	Problem gap
Combine O-PTIR spectroscopy with sample pretreatment (filtration, alcohol) and a support vector machine (SVM) classifier.	Achieved ~91.33% accuracy in distinguishing nylon microplastics from non-nylon MPs.	Focus limited to nylon polymer; generalization to other polymers or environmental samples not fully validated.

- **Source:** Yang, Cihang; Junhao Xie; Aoife Gowen; Jun-Li Xu, et al. “Machine learning driven methodology for enhanced nylon microplastic detection and characterization”, Scientific Reports, Vol. 14, 2024, PP 1–12.

Experiment	Inference	Problem gap
Measured inherent optical properties (absorption, scattering, polarization) of various virgin microplastic particles in seawater (controlled lab conditions).	Different polymers show distinguishable optical signatures; optical detection in situ is a promising route.	Sparse libraries of optical properties; field deployment is complicated by low concentrations, mixtures, background noise, environmental variability.

- **Source: Koestner, Daniel; Robert Foster; Ahmed El-Habashi. “On the Potential for Optical Detection of Microplastics in the Ocean”, Oceanography, Frontiers in Ocean Observing Supplement, 2023, PP 90–97.**

Experiment	Inference	Problem gap
Conducted spectral absorption and angular scattering measurements on microplastic suspensions to build IOP (inherent optical properties) databases.	Provides high-quality optical property data that can help in modeling and differentiating microplastics in water.	Virgin particles used; not accounting for weathering, biofilms or environmental mixing, which can change optical signatures.

- **Source:** Koestner, Daniel; Foster, Robert; El-Habashi, Ahmed. “Measurements of the Inherent Optical Properties of Aqueous Suspensions of Microplastics”, Limnology and Oceanography Letters, Vol. 9, Issue 2, 2024, PP 145–159.

Experiment	Inference	Problem gap
Employed an “electronic tongue” system combining impedance spectroscopy and ML classifiers to detect and classify microplastic types.	The hybrid system was able to classify and detect microplastics based on impedance signatures.	Sensitivity for small particles and performance under mixed matrices (environmental water) may be limited; also, real-time deployment and robustness need further validation. (inferred)

- **Source:** Sarmiento, J.; Velasquez, D.; and Alvarez, L. “Microplastic Identification Using Impedance Spectroscopy and Machine Learning”, Journal of Sensors, Vol. 2024, Article ID 5298635, 2024, PP 1–10.

Experiment	Inference	Problem gap
Used mobile phone microscopy + YOLOv5 deep learning on images of microplastics extracted (via density separation) from consumer product samples (salt, teabags, etc.).	Achieved ~98% accuracy in detecting microplastics using the approach.	Method is limited to extracted samples and controlled settings; may not generalize directly to raw environmental water samples with complex background. (inferred)

- **Source: Md Zayed Bin Zahir Arju; Nafisa Amin Hridi. “Deep-learning enabled rapid and low-cost detection of microplastics in consumer products following on-site extraction and image processing”, Scientific Reports, Vol. 15, 2025, PP 1–14.**

Problem Definition

- Microplastics, being extremely small in size , are difficult to detect and monitor in natural water bodies using conventional visual or manual methods.
- Current detection approaches like FTIR and Raman spectroscopy, though accurate, are expensive, time-consuming, and require skilled operation in controlled laboratory environments.
- There is no affordable, portable, and efficient sensor system available for real-time detection and quantification of microplastics in the field, hindering continuous environmental monitoring and pollution control efforts.

Scope

- Enables rapid, on-site detection and quantification of microplastics in water sources.
- Provides a compact, battery- or solar-powered device suitable for researchers, environmental agencies, and field operators without reliance on laboratory equipment.
- Facilitates the collection, analysis, and visualization of microplastic data, supporting informed decision-making for pollution control and mitigation strategies.

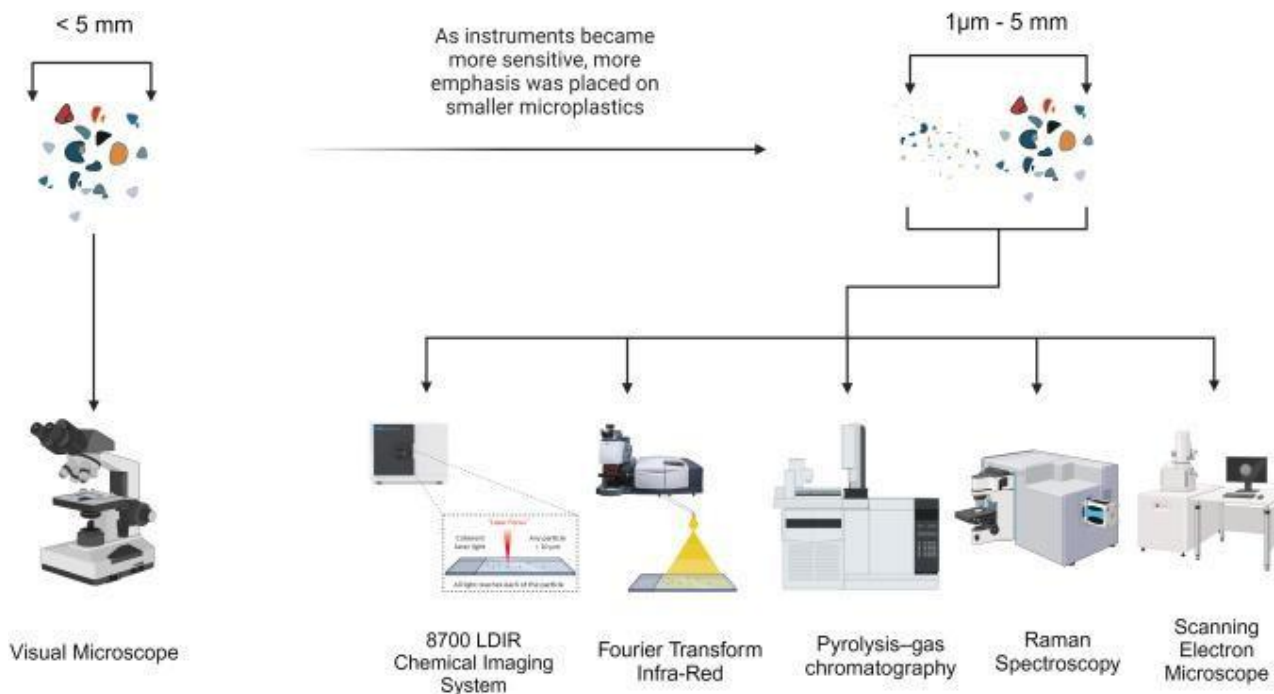
Objectives

- To design and develop a low-cost, portable sensor capable of detecting and quantifying microplastics in various water sources.
- To utilize optical detection techniques such as light scattering, fluorescence, or absorbance for identifying microplastic particles based on their optical properties.
- To integrate intelligent data processing using machine learning algorithms for real-time identification and classification of microplastic types and sizes.
- To create a user-friendly interface for data visualization, logging, and analysis, enabling efficient field deployment and environmental monitoring.

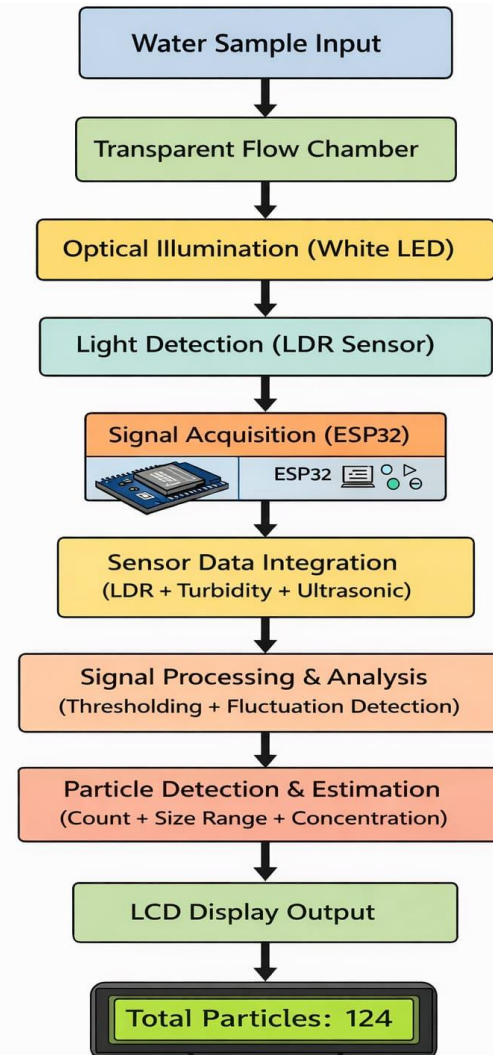
1. **FTIR Spectroscopy:** Identifies microplastics by their unique molecular vibration spectra for polymer-type classification.
2. **Raman Spectroscopy:** Provides precise detection and polymer identification using inelastic scattering of light.
3. **Microscopic Analysis:** Uses optical or electron microscopes to visually observe, size, and count microplastic particles.
4. **Density Separation & Filtration:** Isolates microplastics from water samples using filtration and density-based separation solutions
5. **Limitations:** These methods are expensive, time-consuming, lab-dependent, and unsuitable for rapid, real-time field monitoring.

Source: “Microplastic Identification Using Impedance Spectroscopy and Machine Learning”, Sarmiento, J.; Velasquez, D.; and Alvarez, L, Journal of Sensors, vol 2024, Article ID 5298635, 2024, PP 1–10.

Existing technology of the project



The above figure 1 implies the Current detection approaches though accurate, are expensive, time-consuming, and require skilled operation in controlled laboratory environments. There is no affordable, portable, and efficient sensor system available for real-time detection of microplastics in the field, hindering continuous environmental monitoring and pollution control efforts.



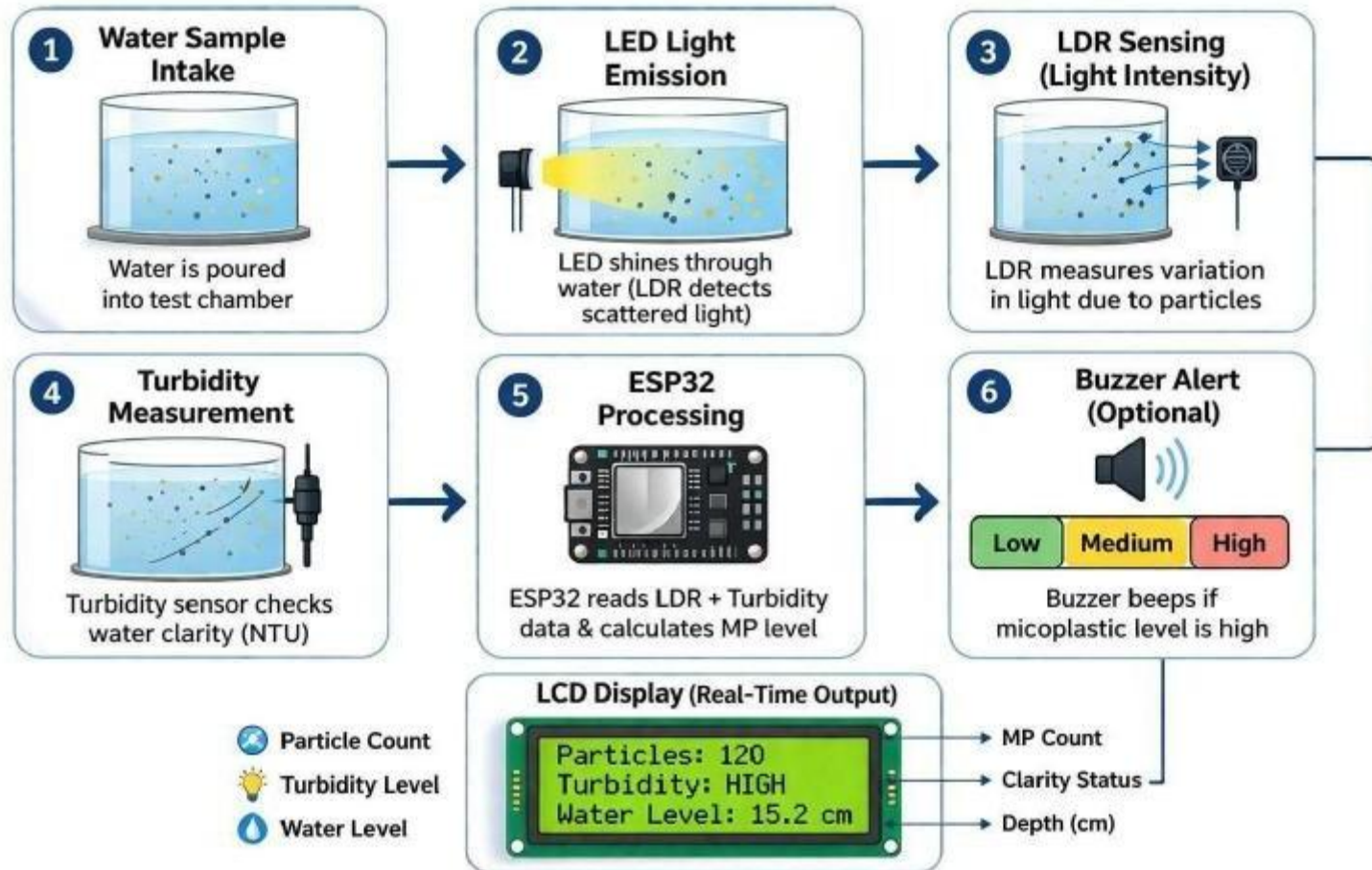


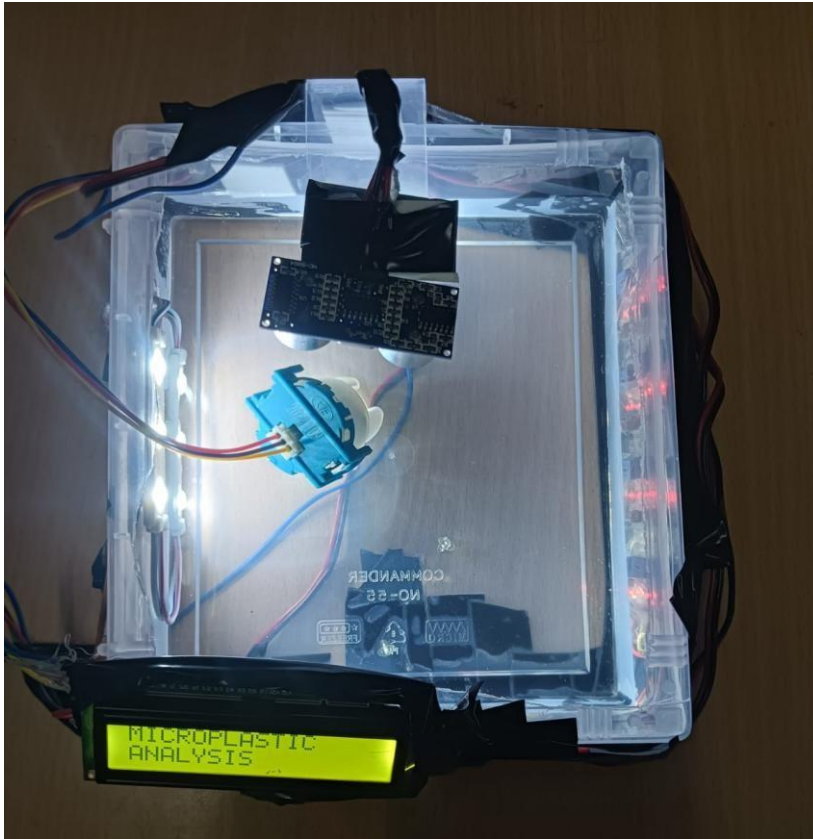
Fig 2 depicts the working process of the microplastics detection prototype using optical sensing and embedded processing



- High feasibility ($\approx 95\%$) due to use of simple and readily available components like LDR, LED, ESP32, and sensors.
- Easy implementation using basic optical sensing and signal processing techniques without complex equipment. Reliable performance achieved through multi-sensor integration (LDR + turbidity + ultrasonic).
- Low cost and low power design makes the system suitable for portable and real-time applications

S.No	Component	Quantity	Cost per Unit (₹)	Total Cost (₹)
1	ESP32 Microcontroller	1	₹300	₹300
2	LDR Sensor	1	₹ 20	₹ 20
3	White LED	2	₹ 10	₹ 20
4	Turbidity Sensor	1	₹ 150	₹ 150
5	Ultrasonic Sensor (HC-SR04)	1	₹ 100	₹ 100
6	LCD Display (16×2)	1	₹ 150	₹ 150
7	Voltage Regulator	1	₹ 30	₹ 30
8	Connecting Wires & Breadboard 1 set		₹ 100	₹ 100
9	Container/Chamber Setup	1	₹ 100	₹ 100
10	Miscellaneous	—	₹ 50	₹ 50
Total Estimated Cost:				₹ 1,020

Total Estimated Cost: ₹ 1,020



1. Prototype Implementation

The developed prototype demonstrates the integration of optical sensing components, including LED illumination, sensor module, and a transparent flow chamber for detecting microplastic particles.

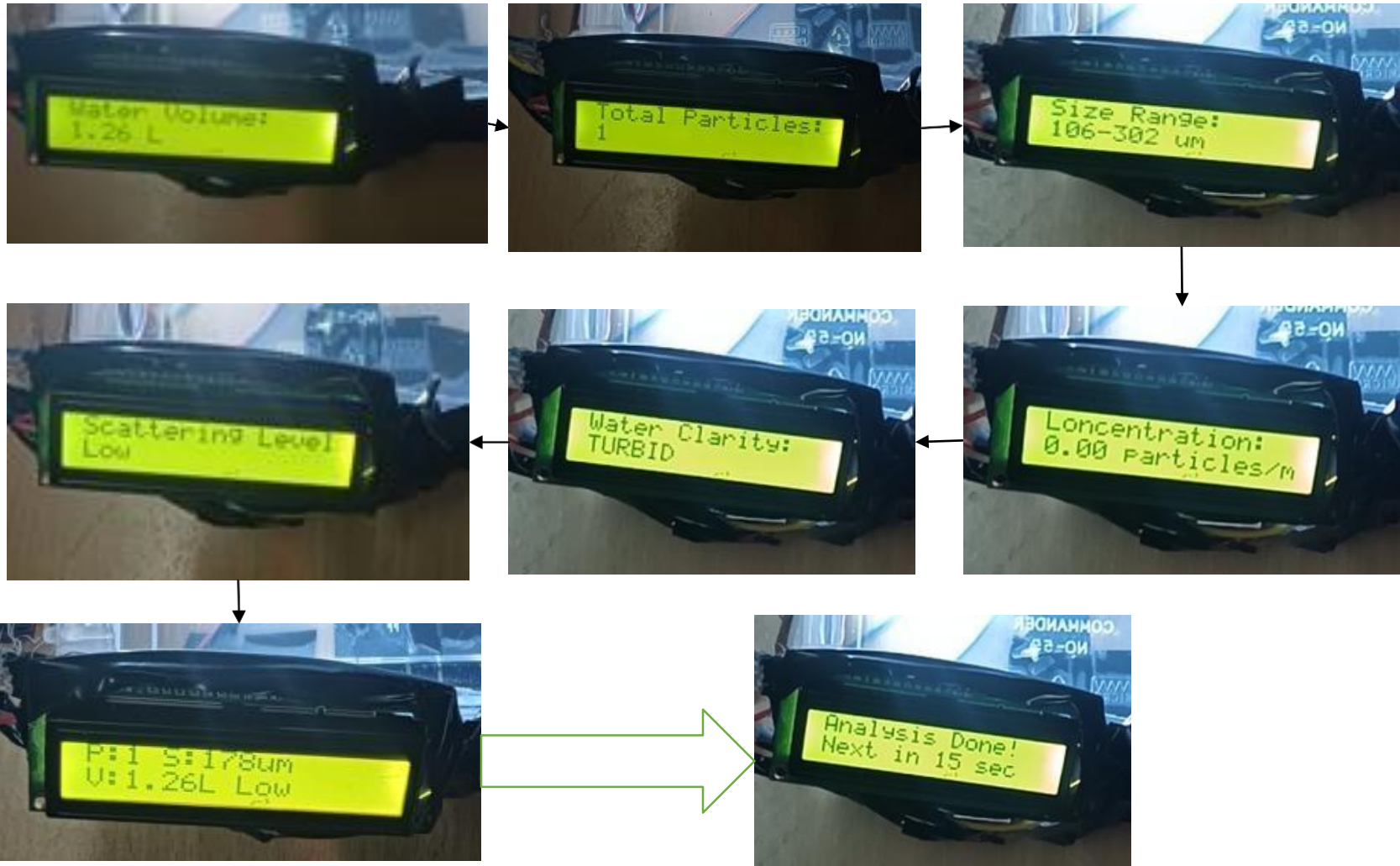
2. Real-Time Display

The system provides real-time analysis results through an LCD display, showing the presence and detection status of microplastic particles in the water sample.

3. Compact and Portable Design

The setup is compact and enclosed within a transparent chamber, making it suitable for portable field applications and easy handling during environmental monitoring.

Results & Discussion



- **Enhanced Detection Range:** The system can be improved to detect microplastics below 100 μm using better optical and imaging techniques.
- **IoT-Based Monitoring:** Integration with IoT can enable real-time data transmission and remote monitoring of water quality.
- **Advanced AI Algorithms:** Use of deep learning models can improve accuracy and enable classification of different types of microplastics.
- **Field Deployment Improvements:** The system can be made more robust and energy-efficient for continuous operation in real environmental conditions.

The proposed low-cost portable microplastic detection system provides an effective and practical solution for identifying microplastic particles in water by integrating optical sensing, image processing, and machine learning techniques. Unlike traditional laboratory-based methods, which are expensive and time-consuming, this system offers real-time detection, portability, and ease of use, making it suitable for field applications such as rivers, lakes, and wastewater monitoring.

The use of affordable components ensures accessibility for research and environmental studies, while the dual detection approach improves reliability and performance. Although the system has certain limitations, such as restricted particle size range and dependence on optical conditions, it establishes a strong foundation for future enhancements. Overall, the project contributes to environmental monitoring and supports sustainable water quality management.



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NOTIFICATION OF ACCEPTANCE

Dear AMRUTHA VARSHINI N,

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The review process for **International Journal of Science, Engineering and Technology (IJSET)** has been completed. Based on the recommendations of the reviewers and the Technical Committee, we are pleased to inform you that your paper has been **ACCEPTED** for online publication.



The developed system of microplastic detection plays an important role in ensuring the achievement of Sustainable Development Goal number 6 (Clean Water and Sanitation), as the development allows carrying out water quality analysis using affordable technologies that will be able to detect pollution caused by the presence of microplastics.

Moreover, the project is relevant in terms of Sustainable Development Goal 12 (Responsible Consumption and Production) because it creates opportunities to make people aware of plastic pollution and promote proper waste management procedures.

1. Yang, Cihang; Junhao Xie; Aoife Gowen; Jun-Li Xu, et al. "Machine learning driven methodology for enhanced nylon microplastic detection and characterization", Scientific Reports, Vol. 14, 2024, PP 1–12.
2. Koestner, Daniel; Robert Foster; Ahmed El-Habashi. "On the Potential for Optical Detection of Microplastics in the Ocean", Oceanography, Frontiers in Ocean Observing Supplement, 2023, PP 90–97.
3. Koestner, Daniel; Foster, Robert; El-Habashi, Ahmed. "Measurements of the Inherent Optical Properties of Aqueous Suspensions of Microplastics", Limnology and Oceanography Letters, Vol. 9, Issue 2, 2024, PP 145–159.---
4. Sarmiento, J.; Velasquez, D.; and Alvarez, L. "Microplastic Identification Using Impedance Spectroscopy and Machine Learning", Journal of Sensors, Vol. 2024, Article ID 5298635, 2024, PP 1–10.
5. Md Zayed Bin Zahir Arju; Nafisa Amin Hridi. "Deep-learning enabled rapid and low-cost detection of microplastics in consumer products following on-site extraction and image processing", Scientific Reports, Vol. 15, 2025, PP 1–14